

# Qiuqi Wang, Ph.D., A.S.A.

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## Academic Appointment

Georgia State University  
– Assistant Professor, Maurice R. Greenberg School of Risk Science  
08/2023 – present

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## Education

Ph.D. Actuarial Science, University of Waterloo  
– Advisor: Ruodu Wang  
06/2023  
M.Phil. Mathematics, Hong Kong University of Science and Technology  
– Advisor: Yue Kuen Kwok  
07/2019  
B.S. Mathematics and Economics, Hong Kong University of Science and Technology  
07/2017  
International Exchange Student, Technical University of Munich  
04/2016 – 09/2016

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## Professional Designation

Associate of the Society of Actuaries (A.S.A.)  
08/2023

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## Research Interests

- Actuarial science; quantitative risk management
  - Operations research; financial engineering
  - Non-parametric hypothesis testing
  - Game theory
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## Publications and Manuscripts

### Pre-publication Manuscripts

- [1] Jiao, Z., **Wang, Q.** and Zhao, Y. (2026). Comparative e-backtests for general risk measures. Preprint.
- [2] Bellini, F., Huang, M., **Wang, Q.** and Wang, R. (2026). Lambda Expected Shortfall. Preprint.

- [3] Chi, Y., Fung, T. C., Peng, L. and **Wang, Q.** (2025). Statistical solutions of optimal multivariate reinsurance under distortion risk measures. Preprint.

### Peer-reviewed Journal Articles

- [1] **Wang, Q.**, Wang, R. and Ziegel, J. (2025). E-backtesting. *Management Science*, available online.
- [2] Han, X., **Wang, Q.**, Wang, R. and Xia, J. (2025). Cash-subadditive risk measures without quasi-convexity. *Mathematics of Operations Research*, available online.
- [3] Boonen, T. J., Chen, Y., Han, X. and **Wang, Q.** (2025). Optimal insurance design with Lambda-Value-at-Risk. *European Journal of Operational Research*, **327**(1), 232–246.
- [4] Aboagye, E., Asimit, V., Fung, T. C., Peng, L. and **Wang, Q.** (2025). A revisit of the optimal excess-of-loss contract. *European Journal of Operational Research*, **322**(1), 341–354.
- [5] Pesenti, S., **Wang, Q.** and Wang, R. (2024). Optimizing distortion riskmetrics with distributional uncertainty. *Mathematical Programming*, **213**, 51–106.
- [6] **Wang, Q.**, Wang, R. and Zitikis, R. (2022). Risk measures induced by efficient insurance contracts. *Insurance: Mathematics and Economics*, **103**, 56–65.
- [7] Embrechts, P., Mao, T., **Wang, Q.** and Wang, R. (2021). Bayes risk, elicibility, and the Expected Shortfall. *Mathematical Finance*, **31**(4), 1190–1217.
- [8] **Wang, Q.**, Wang, R. and Wei, Y. (2020). Distortion riskmetrics on general spaces. *ASTIN Bulletin*, **50**(3), 827–851.
- [9] **Wang, Q.** and Kwok, Y. K. (2020). Real option signaling games of debt financing using equity guarantee swaps. *International Journal of Theoretical and Applied Finance*, **23**(5), 2050036.
- [10] **Wang, Q.** and Kwok, Y. K. (2019). Signaling game models of equity financing under information asymmetry and finite project life. *International Journal of Financial Engineering*, **6**(1), 1950002.

### Dissertations

- [1] Wang, Q. (2023). Characterizing, optimizing and backtesting metrics of risk. *Ph.D. Thesis*. University of Waterloo, Canada.
- [2] Wang, Q. (2019). Real option signaling games under asymmetric information and finite option life. *M.Phil. Thesis*. Hong Kong University of Science and Technology, Hong Kong.

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### Selected Honors and Awards

|                                                                                                                                                                                                                       |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3rd place of Mathematics Doctoral Prize, UW                                                                                                                                                                           | 2024        |
| Statistical Society of Canada's Pierre Robillard Award, SSC                                                                                                                                                           | 2024        |
| <i>An award with one recipient per year recognizing the best Ph.D. thesis defended at a Canadian university in a given year in the fields of probability or statistics</i>                                            |             |
| James C. Hickman Scholar Doctoral Stipend, SOA                                                                                                                                                                        | 2021 – 2023 |
| <i>An award with annually 3-5 recipients currently enrolled in or applying to actuarial science doctoral programs or a related field, in the U.S. or Canada, recognizing great academic performance and potential</i> |             |
| Honorable Mention in ARC Graduate Student Presentation Awards, SOA                                                                                                                                                    | 2022        |
| SAS Chair's Award, UW                                                                                                                                                                                                 | 2021 – 2022 |
| Comprehensive Award (out of 2 recipients), UW                                                                                                                                                                         | 2020        |
| International Doctoral Student Award, UW                                                                                                                                                                              | 2019 – 2023 |
| Statistics and Actuarial Science Doctoral Entrance Award, UW                                                                                                                                                          | 2019        |
| Postgraduate Studentship, HKUST                                                                                                                                                                                       | 2017 – 2019 |
| Scholarship Scheme for Continuing Undergraduate Students, HKUST                                                                                                                                                       | 2015 – 2017 |
| Fong Shu Chuen Scholarship, HKUST                                                                                                                                                                                     | 2015 – 2016 |
| School of Science Dean's List, HKUST                                                                                                                                                                                  | 2014        |

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## Academic Visits

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|------------------------------------------------------------------------------------------------|------------|
| University of New South Wales (School of Risk & Actuarial Studies) (Sydney, Australia)         | 2026       |
| University of Melbourne (Department of Economics) (Melbourne, Australia)                       | 2026       |
| Monash University (Department of Econometrics and Business Statistics) (Clayton, Australia)    | 2026       |
| University of Iowa (Department of Statistics and Actuarial Science) (Iowa City, US)            | 2026       |
| Michigan State University (Department of Statistics and Probability) (Lansing, US)             | 2025       |
| University of Waterloo (Department of Statistics and Actuarial Science) (Waterloo, Canada)     | 2023, 2025 |
| Nankai University (School of Mathematical Sciences) (Tianjin, China)                           | 2023, 2025 |
| Chinese University of Hong Kong-Shenzhen (School of Science and Engineering) (Shenzhen, China) | 2024       |
| Chinese Academy of Sciences (Academy of Math and Systems Science) (Beijing, China)             | 2021, 2023 |
| University of Science and Technology of China (School of Management) (Hefei, China)            | 2021       |

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## Conferences Organized

- 3rd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2022 (Waterloo, Canada)

- 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2021 (Online)

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## Invited Academic Presentations

- 2nd Midwest Actuarial Research Seminars, 2026 (West Lafayette, US)
- China Conference on Insurance and Risk Management, 2025 (Urumqi, China)
- 3rd Foundations and Applications of Decentralized Risk-Sharing, 2025 (Paris, France)
- Young Talents in Actuarial Science and Quantitative Finance, 2025 (Waterloo, Canada)
- Pierre Robillard Invited Address, SSC Annual Meeting, 2024 (St. John's, Canada)
- Young Data Science Researcher Seminar Zurich, 2022 (Online)
- INFORMS Annual Meeting, 2022 (Indianapolis, US)

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## Contributed Academic Presentations

- 1st ASTIN Bulletin Conference, 2026 (Zurich, Switzerland)
- 60th Actuarial Research Conference, 2025 (Toronto, Canada)
- 59th Actuarial Research Conference, 2024 (Murfreesboro, US)
- 27th International Congress of Insurance: Mathematics and Economics, 2024 (Chicago, US)
- 57th Actuarial Research Conference, 2022 (Urbana-Champaign, US)
- 25th International Congress on Insurance: Mathematics and Economics, 2022 (Online)
- CORS/INFORMS International Conference, 2022 (Vancouver, Canada)
- 56th Actuarial Research Conference, 2021 (Online)
- 24th International Congress on Insurance: Mathematics and Economics, 2021 (Online)
- 5th PKU-NUS Annual International Conference on Quantitative Finance and Economics, 2021 (Online)
- 5th Asian Quantitative Finance Seminar, 2020 (Online)
- 1st Waterloo Student Conference in Statistics, Actuarial Science and Finance, 2019 (Waterloo, Canada)
- 23rd Annual International Real Options Conference, 2019 (London, UK)

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## Teaching Experience

### Instructor – Georgia State University

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|--------------------------------------|------------------------------|
| RMI 8300 - Predictive Risk Modeling  | Fall 2024, 2025              |
| AS 3230/8230 - Financial Mathematics | Spring 2024; Fall 2024, 2025 |

### Instructor – University of Waterloo

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|----------------------------------------|-------------|
| ACTSC 371 - Introduction to Investment | Winter 2023 |
|----------------------------------------|-------------|

## Teaching Assistant – University of Waterloo

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|------------------------------------------------------------------------|--------------------------------|
| ACTSC 363 - Casualty and Health Insurance Mathematics 1                | Fall 2022                      |
| ACTSC 964 - Quantitative Risk Management                               | Spring 2022                    |
| ACTSC 446/846 - Mathematics of Financial Markets                       | Winter 2022                    |
| MATBUS 472 - Risk Management                                           | Winter 2021–2022               |
| MATBUS 470 - Derivatives                                               | Fall 2019–2022;<br>Winter 2021 |
| MATBUS 471 - Fixed Income Securities                                   | Fall 2019–2021;<br>Winter 2020 |
| ACTSC 431 - Loss Model I / Casualty and Health Insurance Mathematics 2 | Spring 2020, 2021              |
| ACTSC 445/845 - Quantitative Enterprise Risk Management                | Spring 2020                    |
| ACTSC 231 - Introductory Financial Mathematics                         | Winter 2020                    |

## Teaching Assistant – HKUST

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| MATH 4321 - Game Theory (with tutorials)                                       | Spring 2018, 2019 |
| MATH 4511 - Quantitative Methods for Fixed Income Derivatives (with tutorials) | Fall 2018         |

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## Peer-review Service

- Actuarial: ASTIN Bulletin, Insurance: Mathematics and Economics, Annals of Actuarial Science, North American Actuarial Journal, European Actuarial Journal
  - Finance: Quantitative Finance, Mathematical Finance, Finance & Stochastics
  - OR: Mathematics of Operations Research
  - Probability: Statistics & Probability Letters
  - Economics: Economic Modelling
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